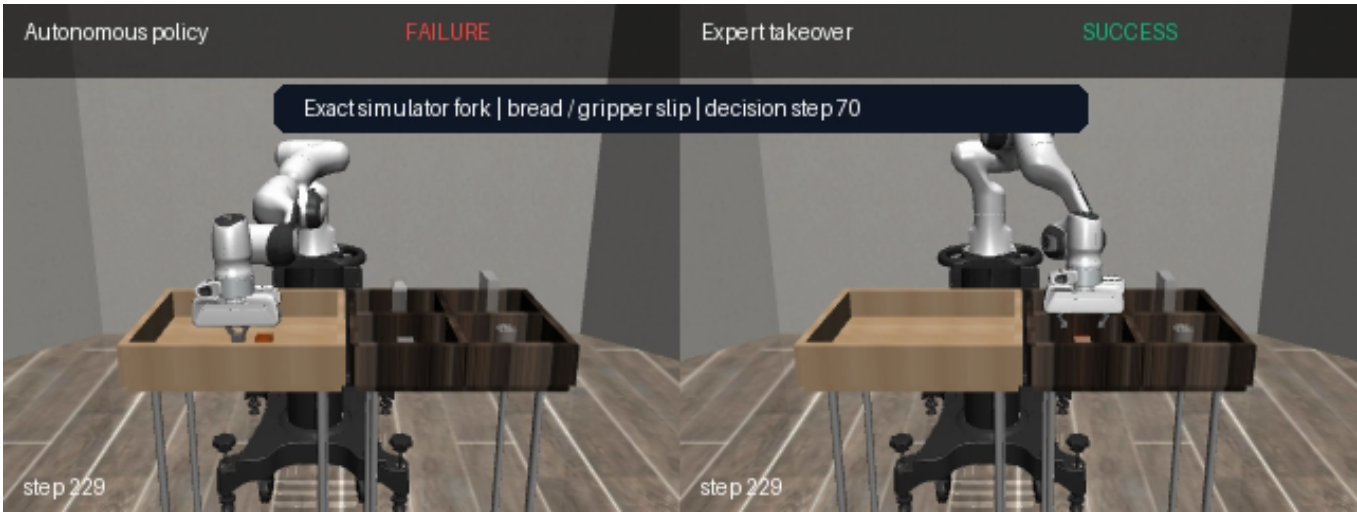


Will Help Change the Outcome?

Learning causal value of expert takeover from exact simulator forks



Exact held-out simulator fork: autonomous continuation fails (left); scripted expert takeover succeeds (right).

0.817	+8.1 pp	5 / 5
held-out value AUROC	value vs. risk near 25%	cross-seed wins over risk

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Simulation-only. Runs locally on Apple Silicon or CPU. No CUDA, cloud compute, paid API, or robot hardware required.

Abstract

Robots that can ask for expert help still need to decide when that help is useful. Failure risk is an incomplete target: a policy can be likely to fail when assistance cannot recover it, or likely to succeed when an unnecessary takeover makes the outcome worse. InterveneSim-Value constructs a causal target by saving the complete simulator and controller state at fixed decision steps and replaying two futures: autonomous continuation and scripted expert takeover. A takeover is helpful only when the assisted branch succeeds and the matched autonomous branch fails. We train a value gate on 1,505 candidates from three policy seeds and evaluate 1,517 candidates from two unseen policy seeds. The value gate reaches 0.817 helpful-intervention AUROC, compared with 0.706 for a temporal failure- risk gate and 0.703 for ensemble uncertainty. Near a 25% intervention budget, value gating achieves 66.2% task success at 23.3% realized intervention, versus 60.4% at 24.6% for risk. An exploratory five-fold policy-seed analysis finds value above risk on all five held-out seeds, with mean advantages of 8.1 points near 25% intervention and 14.7 points near 50%. A secondary frozen-encoder visual probe retains ranking signal across cameras but fails to transfer calibration. The project also contributes deterministic snapshot/restore, a genuine keyboard-takeover data interface, frozen candidate-level results, and a matched counterfactual video. Claims are limited to the simulated system and scripted expert.

1. Research question

Interactive imitation learning can reduce distribution shift by collecting expert actions on states visited by the learner [1, 2]. Budget-aware approaches commonly use novelty, uncertainty, or failure risk to decide when to defer [3]. Those scores answer useful questions, but not the causal deployment question: if the expert takes over now, will the eventual task outcome change?

The distinction matters in both directions. A disturbance may make failure likely after the remaining recovery window has already closed, producing high risk but zero intervention value. Conversely, an unfamiliar-looking state may still be recoverable by the autonomous policy, making uncertainty an expensive false alarm. A takeover can even be harmful if the autonomous branch would have succeeded and the assisted branch fails.

This work asks whether a model trained directly on matched intervention outcomes can produce a better success-intervention frontier than failure risk or ensemble uncertainty. It builds on InterveneSim-X, which found that corrective recovery labels improved disturbed success by 22.8 percentage points over equal-count clean labels across five policy seeds. That earlier study learned from corrections; this study learns when to request them.

2. Exact counterfactual construction

At candidate steps 10, 30, 50, 70, 90, 110, and 130, the environment stores a complete deterministic snapshot. The snapshot contains flattened MuJoCo state, controls, mocap state, environment clocks, termination state, wrapper random-generator state, controller targets, observable caches, and the last raw observation. A restore test verifies equality of the public state and the subsequent transition under a shared action.

Each saved state is replayed into two branches. In the autonomous branch, the learned policy continues and the original disturbance process remains active. In the assisted branch, the scripted expert takes control and synthetic actuator corruption is bypassed, matching the released recovery protocol. Persistent scene state remains. Both terminal outcomes are saved. The binary target is `assisted_success` and not `autonomous_success`; signed value is `assisted_success - autonomous_success`.

This is an exact counterfactual inside the implemented simulator, not a claim of causal identification in the physical world. The expert, takeover semantics, decision grid, and disturbance process are part of the estimand.

3. Experimental protocol

The benchmark uses robosuite [4] Panda PickPlace tasks with can, milk, bread, and cereal objects. Deployment conditions are nominal operation, object shift, Gaussian action noise, multi-step action delay, and gripper slip. Candidate features concatenate the current 35-dimensional task-conditioned state, a causal one-step observation delta, and the autonomous action. The one-step delta resets at episode boundaries.

The primary value model is a two-layer MLP trained with class-weighted binary cross entropy. Training candidates come from policies initialized with seeds 27, 127, and 227. Evaluation candidates come from unseen policies initialized with seeds 327 and 427. Environment reset seeds are also disjoint, and all candidates from an episode remain in the same split.

Three learned gates share the same candidate opportunities. `value_gate` predicts positive matched outcome value. `risk_gate` uses the previously released temporal model that predicts imminent privileged-supervisor takeover. `uncertainty_gate` uses action variance across three independently trained policies. `never_help`, `always_help`, and an earliest-positive-value oracle provide controls.

For target budgets of 10%, 25%, 50%, and 75%, score thresholds are computed only from the maximum candidate score in each training episode. A deployed gate requests help at the first candidate above its threshold. Primary metrics are task success, realized intervention rate, useful-request precision, nominal false alarms, regret to a matched-rate causal oracle, AUROC, and average precision. The full frontier is reported.

4. Primary held-out-policy results

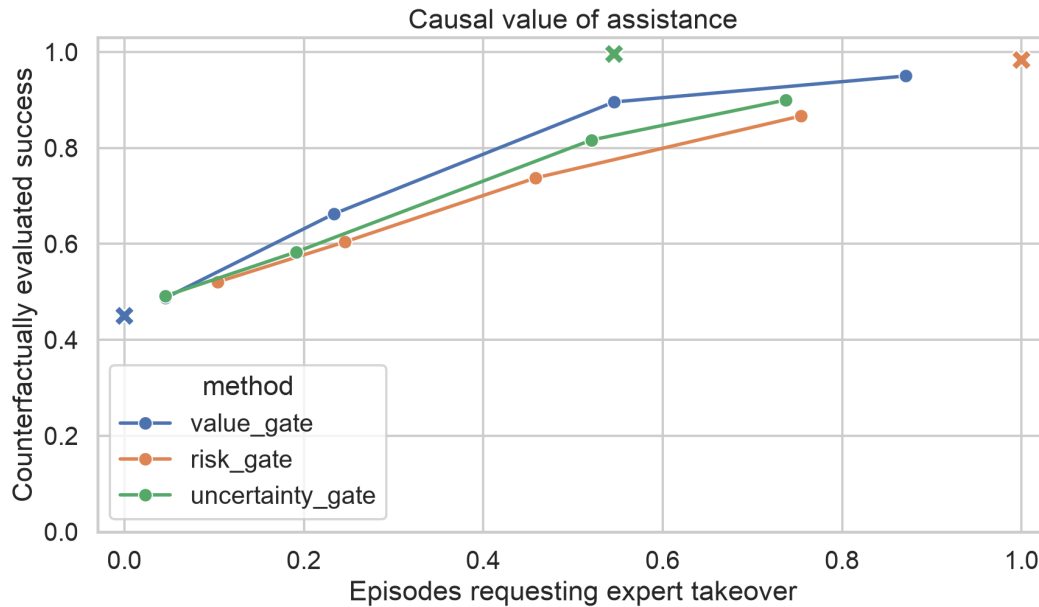


Figure 1. Primary success-intervention frontier on two unseen policy seeds. Crosses show never, always, and causal-oracle controls.

Gate	Target	Success	Intervention	Useful requests	Oracle regret
Value	25%	66.2%	23.3%	91.1%	2.1 pp
Value	50%	89.6%	54.6%	81.7%	10.0 pp
Failure risk	25%	60.4%	24.6%	62.7%	9.2 pp
Failure risk	50%	73.8%	45.8%	62.7%	17.1 pp
Uncertainty	25%	58.3%	19.2%	69.6%	5.8 pp
Uncertainty	50%	81.7%	52.1%	70.4%	15.4 pp

The training split contains 1,505 candidate states from 240 episodes; 41.5% have positive intervention value and 4.3% have negative value. The evaluation split contains 1,517 states from 240 episodes; 45.4% are helpful and 4.4% are harmful. Positive value is concentrated in gripper slip (84.2% of evaluation candidates) and action delay (63.8%), but is not absent from nominal episodes (9.7%). This is why disturbance identity alone is not a sufficient decision rule.

The value gate ranks helpful candidates with AUROC 0.817 and average precision 0.753. Failure risk obtains 0.706 and 0.617; ensemble uncertainty obtains 0.703 and 0.643. At the 25% target budget, value reaches 66.2% success while intervening on 23.3% of episodes. Risk reaches 60.4% at 24.6%, and uncertainty reaches 58.3% at 19.2%. Value requests have 91.1% positive counterfactual precision, compared with 62.7% for risk.

At the 50% target budget, value reaches 89.6% success at 54.6% intervention, compared with 73.8% at 45.8% for risk and 81.7% at 52.1% for uncertainty. These realized rates are not identical, so the figure and tables preserve both axes rather than describing the success gap as a matched-budget treatment effect.

Never asking succeeds on 45.0% of evaluation episodes. Always asking reaches 98.3% but uses the expert on every episode. The causal oracle reaches 99.6% while requesting help on 54.6%. At its 54.6% realized intervention point, the learned value gate has 10.0 points of regret to the matched-rate oracle, leaving meaningful headroom.

5. Exploratory cross-seed robustness

Five-fold policy-seed robustness

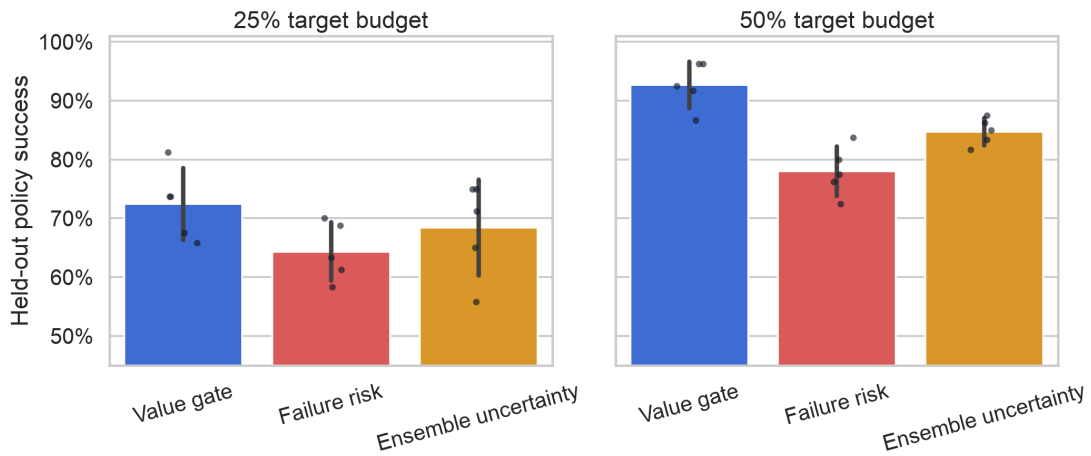


Figure 2. Exploratory five-fold policy-seed audit. Dots are held-out policy seeds; bars show mean and standard deviation.

Target	Reference	Mean value advantage	Std.	Positive seeds
25%	Failure risk	+8.1 pp	4.2 pp	5/5
25%	Uncertainty	+4.0 pp	4.3 pp	4/5
50%	Failure risk	+14.7 pp	3.1 pp	5/5
50%	Uncertainty	+7.9 pp	2.8 pp	5/5

The primary split was fixed before result inspection. After the primary audit, a secondary five-fold leave-one-policy-seed-out analysis was added. Each fold trains value on four policy seeds, evaluates the fifth, retains episode grouping, and derives all thresholds from the four training seeds. Because the analysis was added afterward, it is explicitly exploratory.

Near the 25% target budget, value averages 72.4% success at 25.5% intervention; risk averages 64.3% at 25.1%, and uncertainty averages 68.4% at 25.6%. The paired value-minus-risk difference is +8.1 points with a 4.2-point standard deviation and is positive on all five held-out policy seeds. Value exceeds uncertainty on four of five seeds by 4.0 points on average.

Near the 50% target, value averages 92.7% success at 53.9% intervention. Risk averages 78.0% at 51.4% and uncertainty averages 84.7% at 49.9%. Value beats both on every held-out seed: +14.7 +/- 3.1 points over risk and +7.9 +/- 2.8 over uncertainty. The small number of policy seeds supports consistency and effect-size reporting, not asymptotic significance claims.

6. Secondary visual and human interfaces

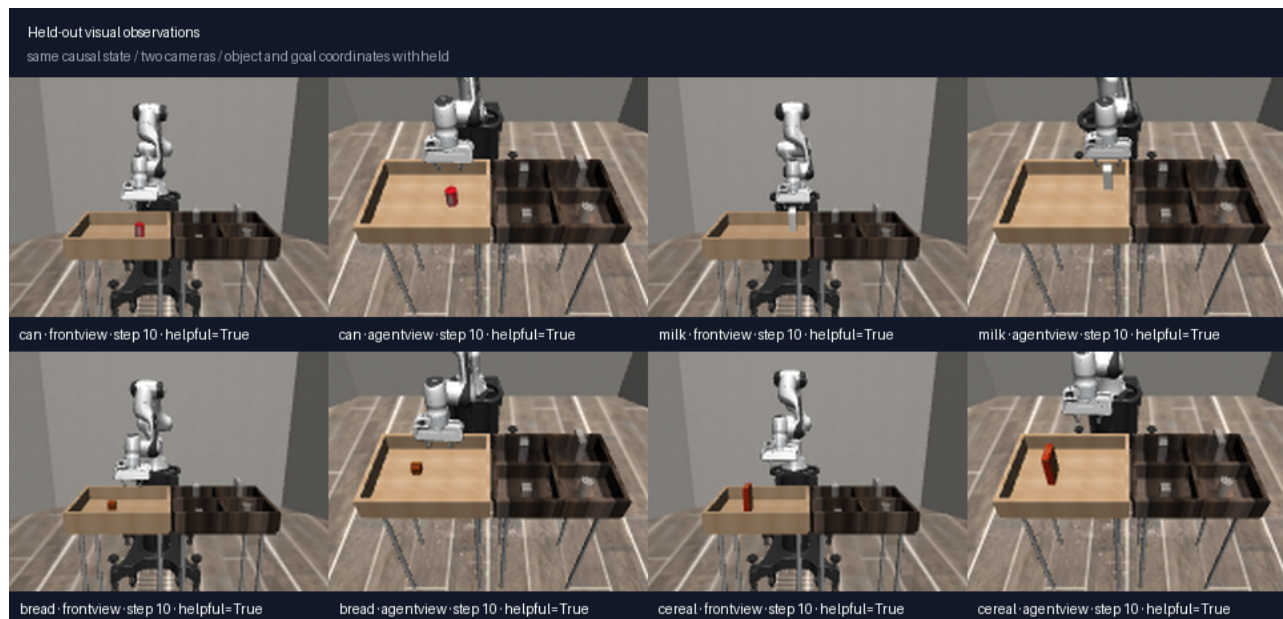


Figure 3. Matched held-out-policy states from front-view and zero-shot agent-view cameras.

Camera	AUROC	Average precision	Recall @ 0.5	Positive rate
frontview	0.712	0.582	80.6%	59.4%
agentview	0.696	0.586	1.9%	1.2%

The primary result uses simulator state to isolate the causal question. A secondary visual pilot tests whether the label can support perception research. A frozen ImageNet ResNet-18 [5] encodes 160 x 120 RGB observations. A small learned head combines its 512 features with robot state, control progress, and task identity while withholding object and goal coordinates. Training uses 249 front-view images; evaluation uses 512 images from an unseen policy seed across front-view and agent-view cameras.

Front-view AUROC is 0.712 with 80.6% recall at threshold 0.5. Zero-shot agent-view AUROC is 0.696, indicating that some ranking signal transfers. Calibration does not: only 1.2% of agent-view candidates score above 0.5 and recall falls to 1.9%. This is a useful negative result and not evidence of a deployable visual gate. Lighting and textures were not varied.

The release also includes capture-human-corrections. A learned policy controls the simulator until a person presses T, after which keyboard Cartesian and gripper commands are recorded with the exact takeover step and a per-action human mask. Human archives use a distinct schema and are never silently mixed with scripted data. No participants were recruited and no human-performance result is claimed.

7. Threats to validity

The largest limitation is external validity. MuJoCo state restore makes paired outcomes exact for the implemented dynamics, but physical systems cannot generally be rewound. The scripted expert has privileged geometry and deterministic logic. Assisted branches bypass synthetic actuator corruption, which represents a reliable expert control channel rather than a shared hardware failure. Different experts or takeover channels define different values.

Candidate decisions occur on a fixed grid rather than continuously. Episode-level budgets are targeted through training score quantiles, so realized evaluation rates can differ. Primary uncertainty is also limited by five total policy initializations. The post-audit cross-validation reuses all seeds across different folds and is therefore not independent replication.

The visual dataset is small and uses a pretrained encoder. It probes representation and calibration, not end-to-end visual control. No result establishes real-robot safety, sim-to-real transfer, human intervention quality, or robustness to lighting and texture.

8. Reproducibility

The entire pipeline runs locally on an Apple M4 Pro with 64 GB unified memory using PyTorch MPS; CPU is supported and CUDA is not required. All software is free, dependencies are locked, and no cloud service, paid API, or physical robot is used. Collection is sharded by policy seed and task. Datasets, model checkpoints, and completed folds are cached so an interrupted run resumes without discarding completed work.

The public artifact freezes candidate-level branch outcomes, gate episode records, resolved configurations, thresholds, metrics, cross-seed tables, plots, a matched counterfactual video, model and dataset cards, and tests that assert the reported primary and exploratory claims. Large observation tensors and checkpoints are regenerated locally.

9. Conclusion

Failure risk and uncertainty are not the same quantity as intervention value. In this benchmark, learning directly from matched autonomous and assisted futures produces a better held-out-policy success-intervention tradeoff, more precise requests, and consistent advantages across five exploratory leave-one-policy-seed-out folds. Exact simulation forks make the causal target inspectable; the visual calibration failure and strict claim boundaries keep the result honest. InterveneSim-Value is a compact platform for studying not only how robots recover from correction, but when they should ask for it.

References

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